

Research

Hybrid CNN-LSTM model for ECG-based arrhythmia detection in internet of medical things

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Abstract

Electrocardiogram (ECG) signals play a crucial role in the internet of medical things (IoMT) by enabling real-time heart health assessment, particularly for cardiovascular disease monitoring and personalized health management in sports and fitness applications. This study proposes a hybrid deep learning model that integrates a long short-term memory (LSTM) network and a convolutional neural network (CNN) to enhance ECG signal classification. The LSTM network captures the temporal dependencies of ECG sequences, while the CNN extracts localized spatial features, allowing for a comprehensive analysis of heart rhythms. By utilizing preprocessed ECG data as input, our model effectively combines feature extraction and classification into a unified framework. We evaluate the proposed approach using a publicly available dataset. Experimental results demonstrate that our method achieves high accuracy in distinguishing various heart rhythm patterns, offering a promising solution for automated arrhythmia detection in IoMT applications.

Keywords IoMT · Medical health · ECG signal · Arrhythmia classification · Neural networks

1 Introduction

The emergence of smart sports has garnered increasing attention worldwide, particularly in applications that enhance quality of life through physical activity monitoring and health promotion. By integrating smart technology with personalized health management, real-time physiological monitoring has become an essential tool for improving human function, preventing chronic diseases, and providing timely medical interventions. Among these, cardiovascular diseases (CVDs) pose a significant threat due to their unpredictable onset and severe consequences. The use of wearable devices for real-time heart monitoring offers a proactive approach to detecting abnormalities and facilitating timely medical responses. These advancements fall under the domain of the Internet of Medical Things (IoMT), a specialized subset of the internet of things (IoT) that integrates medical sensors, wearable devices, and AI-driven analytics to enable continuous patient monitoring and intelligent healthcare solutions. Unlike general IoT applications, which encompass diverse domains such as smart homes and industrial automation, IoMT is designed specifically for medical environments, ensuring real-time data processing, patient safety, and regulatory compliance [1, 2]. Recent research has demonstrated the effectiveness of IoMT in healthcare applications, such as stress recognition in smart home environments [1] and cyberthreat detection and data authentication in medical systems [2]. Continuous ECG monitoring within IoMT frameworks plays a crucial role in preventing, diagnosing, and managing cardiovascular conditions, where early detection of arrhythmias and other irregular heart rhythms can significantly reduce complications and improve patient outcomes [3, 4].

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Electrocardiography (ECG) is the standard technique for assessing heart health by recording the electrical activity of the heart over time. Given the growing emphasis on personalized health monitoring and the disparity in medical resources, automated arrhythmia detection systems have become a research focus. The American Association for the Advancement of Medical Instrumentation (AAMI) classifies ECG signals into five heartbeat categories: non-ectopic (N), supraventricular (S), ventricular (V), fusion (F), and unknown (Q) beats [5]. Accurate classification of these beats is critical for early diagnosis and effective treatment of cardiac disorders.

Traditional machine learning methods have been widely applied to ECG classification, including support vector machines (SVM) [6], artificial neural networks (ANN) [7], and decision trees (DT) [8]. However, these approaches require manual feature extraction, which is often complex, time-consuming, and susceptible to human bias. Furthermore, traditional methods struggle with capturing the intricate nonlinear dependencies in ECG signals, potentially leading to suboptimal classification performance. Given the short duration and low amplitude of ECG signals, effective feature extraction remains a significant challenge.

In contrast, deep learning-based approaches have demonstrated superior performance in automatic arrhythmia detection. Convolutional Neural Networks (CNNs) are particularly effective in extracting local morphological patterns from ECG signals, while Long Short-Term Memory (LSTM) networks excel at capturing long-term temporal dependencies [9]. CNN-based models have been successfully applied in various ECG classification studies. For example, prior research has utilized wavelet transforms and Fourier transforms to convert ECG signals into two-dimensional spectrograms for CNN-based classification, achieving accuracy rates above 98% [10]. Similarly, deep CNN models incorporating Focal loss functions have addressed data imbalance issues in ECG datasets [11]. Additionally, LSTM-based models have demonstrated their ability to capture temporal dependencies within ECG sequences, enhancing classification accuracy for arrhythmias [12].

While CNNs and LSTMs have been individually employed for ECG analysis, few studies have effectively combined these architectures into a unified framework that optimally integrates spatial and temporal feature extraction. To address this gap, this study proposes a novel hybrid deep learning model that synergistically integrates CNN and LSTM networks with an attention mechanism to improve ECG classification performance. Unlike prior work that typically employs either CNN or LSTM in isolation, our method leverages their complementary strengths to improve classification accuracy and robustness. The incorporation of an attention mechanism further enhances our model by selectively focusing on the most informative time steps, ensuring that critical ECG features are retained while reducing the impact of less relevant signal components.

Beyond methodological innovation, we have also strengthened our discussion on practical applicability and the broader implications of our work. We now emphasize that our end-to-end approach eliminates the need for handcrafted feature engineering, enabling fully automated ECG classification. This not only enhances scalability but also facilitates real-time applications in wearable medical devices and IoMT-based health monitoring. Additionally, to address the common challenge of class imbalance in ECG datasets, we employ an effective data augmentation strategy based on a sliding window approach, which significantly improves model generalization across different heartbeat classes.

To further support the novelty of our approach, we have conducted extensive experimental validation using the MIT-BIH arrhythmia database, comparing our method against multiple state-of-the-art algorithms. The results demonstrate that our model consistently outperforms existing techniques, achieving higher classification accuracy and robustness across different arrhythmia classes. The main contributions of this paper can be summarized as follows:

- (1) A new system model is introduced, combining LSTM and CNN in series with an attention mechanism to automatically classify arrhythmias from ECG signals.
- (2) The proposed method utilizes the publicly available MIT-BIH database for validation.
- (3) Comparative experiments were conducted using several evaluation metrics to assess the performance of the proposed algorithm. The results demonstrate that the method significantly outperforms other approaches.

2 Related work

The advancement of information and artificial intelligence technologies has significantly improved the automatic classification of ECG signals. A common class of methods are classification algorithms based on waveform features. The classification algorithm based on waveform features mainly includes three parts: signal preprocessing, feature extraction and classification and identification. ECG generally contains multiple heartbeats. A heart beat is composed of P wave, QRS complex, T wave, etc. The extraction of ECG information features based on waveform shape requires accurate detection of these waveforms. Common waveform detection algorithms include wavelet transform [13], template matching method

[14], pattern recognition method [15], energy threshold method [16], etc. At present, the detection method based on R wave has a relatively high accuracy, while the amplitude of P wave and T wave is low, and the feature extraction algorithm remains underdeveloped. The ECG feature extraction method also includes Wavelet transform method [17], power spectrum feature [18], higher order statistics method [19], principal component analysis [20], etc. Kunyang Li et al. [21] combined wavelet transform and morphology to detect the QRS complex. Feature points, by parameters such as QRS wave width and RR interval, combined with medical knowledge to classify heart beats. The accuracy rate reached 94.2%. Li et al. [22] employed a genetic algorithm-based least squares double support vector machine, achieving classification of five arrhythmia types with an accuracy of 99.14%. Despite the success of the aforementioned ECG signal classification systems, several challenges remain due to the unique characteristics of ECG signals: (1) ECG waveforms and patterns vary across different patients and even within the same patient over time, making it difficult to rely on consistent ECG features for evaluation; (2) Manual feature selection is influenced by numerous factors, requiring significant time and cost; (3) The training efficiency of shallow neural networks and certain machine learning methods is limited, hindering their practical application in clinical settings.

Deep learning methods have the ability to automatically extract features through hierarchical structures, eliminating the need for manual feature selection. In recent years, Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs) have demonstrated remarkable success across a wide range of applications [23]. These techniques have also been increasingly adopted in arrhythmia detection research, where they excel in automating the classification process. For instance, Meng et al. [24] utilized raw ECG signals and RR intervals as inputs and employed a 4-layer Deep Belief Network (DBN) to classify six types of arrhythmias, achieving an impressive overall accuracy of 98.49%. Such approaches showcase the effectiveness of deep learning in handling complex physiological data, enabling more accurate and efficient arrhythmia detection. Acharya et al. [25] used a 9-layer CNN model to realize the identification and classification of five kinds of arrhythmias, and obtained a classification accuracy of 94.03%. Judging from the above literature, the ECG automatic classification technology based on deep learning has outstanding advantages over traditional methods. However, the learning process of deep neural networks is relatively complex, the model calculation is large, and large labeled data samples are required to train the network, which is expensive in time and hardware.

3 The proposed methodology

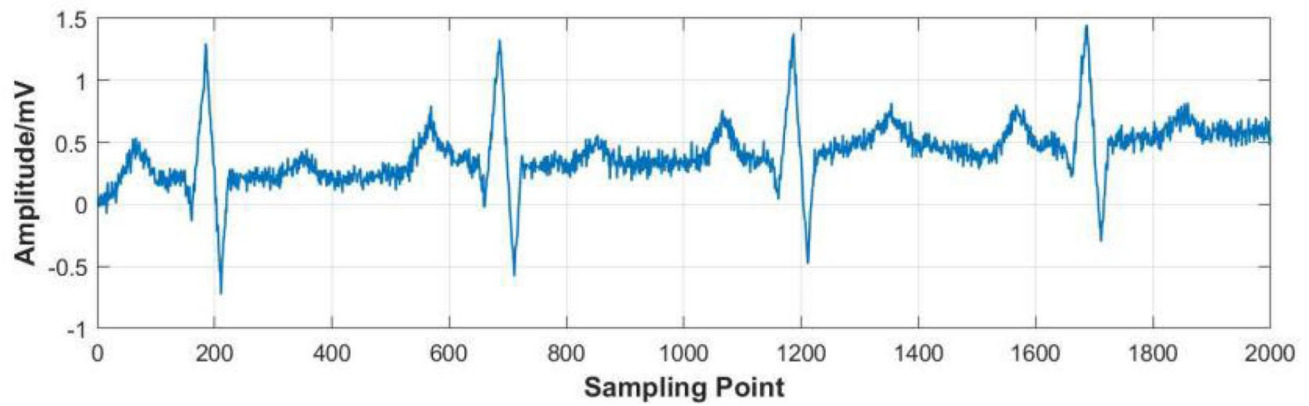
3.1 Data preprocessing

During ECG signal acquisition, various types of noise, such as baseline drift, power frequency interference, and EMG disturbance, often become embedded in the signal. These noise factors can compromise the accuracy of ECG classification. To mitigate this issue and enhance the signal-to-noise ratio of the ECG signal, it is necessary to perform detrending, filtering and other operations before sending the data into the neural network. As shown in Fig. 1a, the original ECG signal shows a linear trend, and the effect of removing the trend is shown in Fig. 1b. It can be seen that the baseline of the signal no longer drifts. Then, the wavelet method is used to denoise the ECG signal. In wavelet coefficient analysis, coefficients with smaller absolute values are set to zero, while those with larger absolute values are either retained or shrunk. The processed wavelet coefficients are then reconstructed to obtain a de-noised signal. As shown in Fig. 1c, the wavelet transform effectively reduces noise while preserving the useful signal.

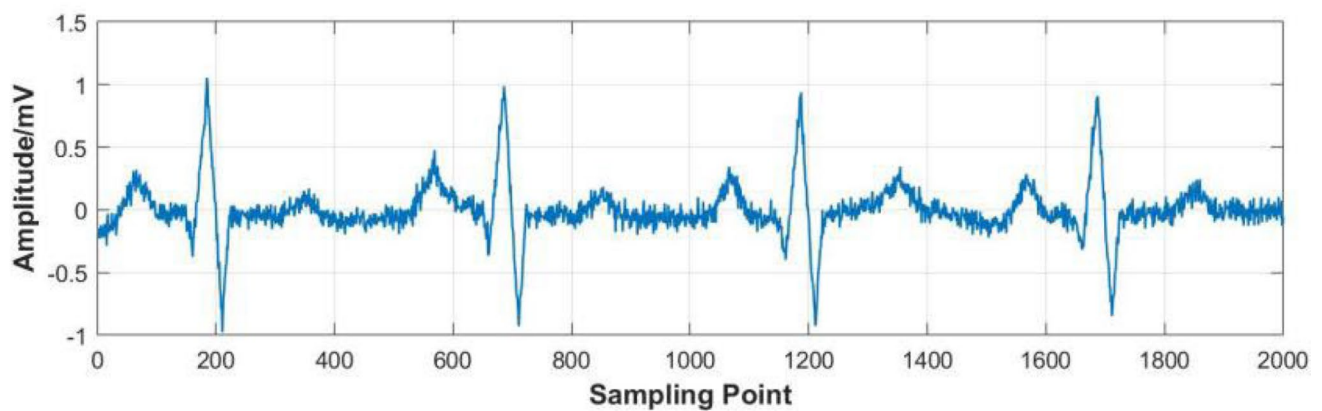
3.2 CNN architecture for ECG feature extraction

This paper proposes a convolutional neural network (CNN) to automatically extract features from ECG signals, transforming the input into a 1-dimensional feature vector of fixed length and reduced size. This feature vector is then fed into a long short-term memory (LSTM) network for training and testing, enabling automatic arrhythmia classification. CNN effectively captures data features through its convolution operations and hierarchical structure, classifying input information accordingly. The convolution kernel repeatedly operates across receptive fields in the input, generating feature maps, which are then processed by the pooling layer for feature selection and information filtering. By summarizing adjacent output statistics, the pooling layer reduces the computational load.

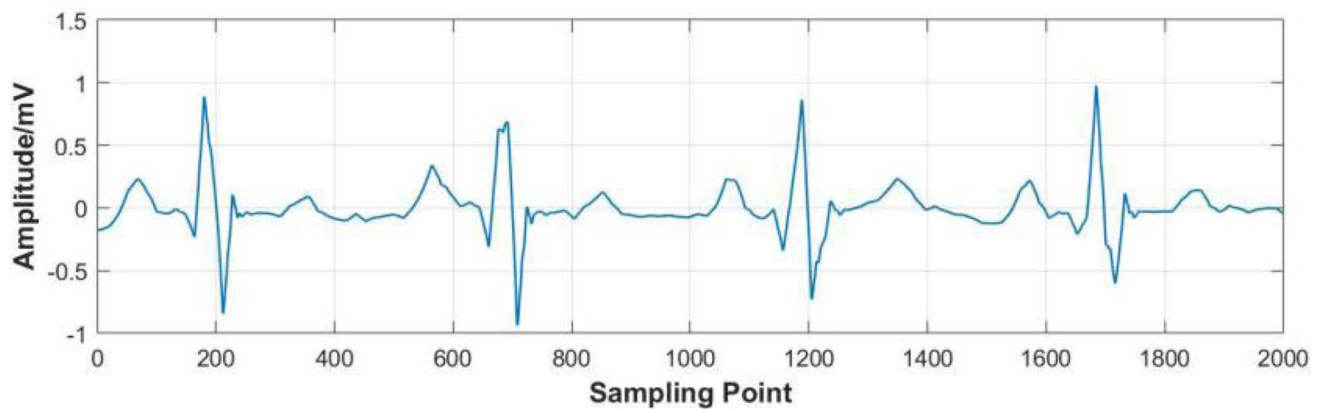
In this study, the CNN architecture is inspired by the VGGNet design, adapted specifically for one-dimensional ECG signals. A 1D convolutional framework is employed to efficiently extract features from ECG signals, allowing the model to focus on relevant heart rhythm patterns that are characteristic of normal and abnormal heartbeats. The convolutional



(a) Original ECG Signals



(b) Detrended ECG Signals



(c) ECG Signals After Wavelet Denoising

Fig. 1 ECG signals preprocessing

layers are designed to capture local morphological characteristics and short-term dependencies within the time-series data. This enables the extraction of high-level features, such as the morphology and dynamics of heartbeats, which are critical for accurate arrhythmia detection.

3.2.1 Convolutional layer and feature extraction

Each convolutional layer consists of a set of trainable filters (also known as kernels) that systematically scan through the input ECG sequence to extract relevant local features. Specifically, the output of a neuron in the convolutional layer is computed as:

$$y_i = f\left(\sum_{j=1}^m w_j \cdot x_{i-j_m} + b\right)$$

where x_t ($t = 1, 2, 3, \dots, n$) represents the input ECG sequence, w_j denotes the weight matrix of the convolution kernel, b is the bias term, and m is the number of convolution filters. The function $f(\cdot)$ is the activation function that introduces non-linearity to enhance the expressiveness of the network.

The 1D convolution is particularly suited for ECG signal analysis as it effectively captures local temporal dependencies, such as QRS complex variations, waveform morphology, and amplitude fluctuations. By scanning over the signal with overlapping receptive fields, the model learns local features that are highly discriminative for arrhythmia classification.

3.2.2 Batch normalization and activation function

To enhance the model's stability and improve generalization, Batch Normalization (BN) is applied after each convolutional layer. BN helps mitigate the internal covariance shift problem by normalizing each mini-batch, which accelerates convergence and reduces the risk of overfitting. The batch normalization transformation is given by:

$$\hat{x}_i = \frac{x_i - \mu}{\sqrt{\sigma^2 + \epsilon}} \cdot \gamma + \beta$$

where μ and σ^2 are the mean and variance of the input batch, γ and β are learnable scaling parameters, and ϵ is a small constant added for numerical stability.

For non-linearity, we employ the Rectified Linear Unit (ReLU) activation function. ReLU is chosen due to its ability to mitigate the vanishing gradient problem, allowing for more effective training in deep networks. Furthermore, it introduces sparsity in activations, reducing unnecessary computations and improving computational efficiency.

3.2.3 Pooling and dimensionality reduction

After feature extraction by the convolutional layers, pooling operations are applied to reduce the dimensionality of the feature maps while retaining essential information. Max pooling is commonly used to preserve the most salient features by selecting the highest value within each local receptive field. The reduced feature maps obtained from pooling layers ensure computational efficiency and robustness to small signal variations.

3.2.4 Feature map transformation and output

Once the hierarchical convolutional layers have extracted informative representations from the ECG signals, the resulting feature maps are passed through a final convolutional layer that produces a fixed-length feature vector. This step ensures that the extracted features maintain discriminative power while reducing redundant information.

The final output of the CNN is then fed into the LSTM network, which is responsible for modeling long-term dependencies in the ECG sequences. The combination of CNN for spatial feature extraction and LSTM for temporal sequence modeling allows the model to fully capture both morphological and sequential patterns in ECG signals, leading to improved classification performance.

3.3 LSTM-attention layer

Long short-term memory (LSTM), an advanced version of the recurrent neural network (RNN), is employed to handle the temporal dependencies in the ECG signal. LSTM addresses key challenges such as the vanishing gradient problem, enabling it to effectively capture long-term sequential patterns in the data. This capability is particularly important for ECG signals, where the classification of heartbeats is often dependent on context over extended time periods. In our methodology, the LSTM layer receives the output feature vector from the CNN and learns to capture the temporal dependencies inherent in the ECG signal. The LSTM's gating mechanisms—input, forget, and output gates—allow it to retain critical contextual information over long sequences. This ability to retain memory is crucial for ECG classification, where the context of previous heartbeats can greatly influence the classification of the current heartbeat.

LSTM networks are designed to address critical challenges that traditional RNNs face, such as the vanishing gradient and exploding gradient problems, by introducing special gating mechanisms. These gates—input gate (i), forget gate (f), and output gate (o)—play a crucial role in regulating the flow of information across the network and ensuring that important information is retained over long sequences.

Input gate (i): The input gate controls the extent to which the current input X_t is incorporated into the cell state (C_t). It determines how much of the incoming data is relevant to the ongoing learning process and should be stored in memory.

Forget gate (f): The forget gate regulates the retention of information from the previous time step C_{t-1} . It decides which parts of the previous memory should be discarded, allowing the network to forget irrelevant past information.

Output gate (o): The output gate determines how much of the cell state C_t contributes to the current output H_t . It decides which internal state information should be passed forward to the next layer or the next time step.

The cell state C_t , which serves as the memory of the network, evolves over time by interacting with the gating mechanisms. These gates are controlled by sigmoid and tanh activation functions. When the input ECG signal vector matrix is $X = [x_1, x_2, \dots, x_k]$, the update formula of LSTM is:

$$i_t = \sigma(W_i \cdot [H_{t-1}, x_t] + b_i)$$

$$o_t = \sigma(W_o \cdot [H_{t-1}, x_t] + b_o)$$

$$f_t = \sigma(W_f \cdot [H_{t-1}, x_t] + b_f)$$

$$C_t = f_t \otimes C_{t-1} + i_t \otimes \tanh(W_c \cdot [H_{t-1}, x_t] + b_c)$$

$$H_t = o_t \otimes \tanh(C_t)$$

In the formula, $\sigma(\cdot)$ represents the Sigmoid activation function, $\tanh(\cdot)$ is the hyperbolic tangent function, W denotes the weight matrix, b is the bias term, and H_t is the final output. The output H_t , which encapsulates the context information extracted by LSTM, serves as the input for multi-channel attention.

The cell state C_t maintains a linear relationship with the previous cell state C_{t-1} , enabling LSTM to retain long-term dependencies. When the forget gate is active (i.e., when its output is close to 1), the network effectively "remembers" the previous state and can incorporate this memory with the new input. This mechanism allows the network to capture relevant past information over extended periods, which is especially important for ECG signals where contextual history is critical for detecting arrhythmias.

The output H_t , which contains the aggregated context of the entire sequence up to time step t , is passed through an attention mechanism to further refine and focus on the most relevant features for classification. Attention mechanisms allow the model to weigh the importance of each time step or feature in the sequence, prioritizing information that is most significant for the final classification task. In our methodology, the output H_t from the LSTM is used as the input for a multi-channel attention mechanism. The attention layer assigns different attention scores to different time steps or segments of the ECG signal, thereby emphasizing critical portions of the sequence and minimizing the impact of less relevant parts. This mechanism helps the model to focus on the contextual relationship between heartbeats over long sequences, improving the accuracy of arrhythmia classification.

3.4 Neural network model combining LSTM and CNN

The proposed neural network model integrates a 15-layer one-dimensional convolutional neural network (1D-CNN) with a long short-term memory (LSTM) network to efficiently capture both spatial and temporal characteristics of ECG signals, enabling robust and accurate arrhythmia classification. The CNN component is responsible for extracting high-level spatial representations from raw ECG signals, focusing on local morphological features such as P-waves, QRS complexes, and T-waves, which are essential for distinguishing normal and abnormal heart rhythms. By leveraging multiple convolutional layers with progressively increasing filter numbers—from 16 to 32 and eventually to 64—the network is able to extract multi-scale features, preserving fine-grained details while ensuring hierarchical feature abstraction. To further enhance stability and prevent overfitting, batch normalization (BN) is applied after each convolutional layer, normalizing the feature distributions across mini-batches. The Rectified Linear Unit (ReLU) activation function is employed to introduce non-linearity, facilitating the learning of complex patterns within ECG sequences. After feature extraction, the CNN output is transformed into a 38-dimensional feature vector, which serves as input to the subsequent LSTM network.

The LSTM module is specifically designed to model the temporal dependencies inherent in ECG signals, ensuring that contextual information from previous heartbeats is retained to improve classification performance. Unlike traditional machine learning models that process individual beats independently, the LSTM layer effectively captures sequential patterns, making it well-suited for handling the long-term dependencies present in physiological signals. The network is configured with 32 LSTM units, incorporating gating mechanisms such as input, forget, and output gates to regulate information flow and mitigate vanishing gradient issues. The memory cell state is updated through a combination of past information retention and new input integration, allowing the model to selectively store relevant cardiac patterns over extended time periods. Moreover, the combination of CNN and LSTM eliminates the need for manual feature engineering, enabling an end-to-end learning framework that directly maps raw ECG signals to classification outcomes. The final output from the LSTM is fed into a fully connected layer, followed by a Softmax function that computes the probability distribution across different arrhythmia classes. This hybrid deep learning architecture ensures a high level of feature expressiveness while maintaining computational efficiency, making it suitable for real-time ECG analysis in Internet of Medical Things (IoMT) applications. The overall network structure, as illustrated in Fig. 2, demonstrates a well-balanced approach to feature extraction, sequence modeling, and classification, ensuring a comprehensive understanding of ECG signal dynamics. As shown in our experiments of the following section, CNN significantly reduces training time by compressing ECG signals from 300 points to 38 feature vectors, achieving comparable accuracy to standalone LSTM but with $5.8 \times$ faster training speed.

3.5 Data augmentation

Deep learning models exhibit strong feature extraction and pattern recognition capabilities; however, they are inherently prone to overfitting when trained on imbalanced datasets, particularly in medical applications such as ECG classification. In this study, the dataset presents a significant class imbalance, with certain heartbeat categories, such as Supraventricular (S) and Fusion (F) beats, being underrepresented compared to the Normal (N) class. This imbalance can lead to biased model learning, where the classifier becomes skewed toward majority classes, thereby reducing sensitivity in detecting minority class arrhythmias. To address this issue and improve the generalization ability of the model, data augmentation techniques are employed to artificially expand the training dataset. In computer vision tasks, common data augmentation methods include mirroring, rotation, scale transformation, cropping, and noise perturbation. However, due to the time-series nature of ECG signals, these conventional augmentation strategies are not directly applicable. Instead, we adopt the sliding window method, which generates additional training samples while preserving the temporal dependencies and morphological structures of ECG waveforms. Specifically, for each heartbeat segment, multiple overlapping training samples are created by systematically shifting the window's starting position. This technique effectively increases the diversity of training samples, thereby enhancing the model's ability to recognize heartbeat variations.

To further mitigate class imbalance, the sliding window augmentation is selectively applied to minority classes within the training set. By increasing the number of samples for S and F categories, the overall class distribution is more balanced, improving the model's capability to distinguish between different arrhythmic patterns. While

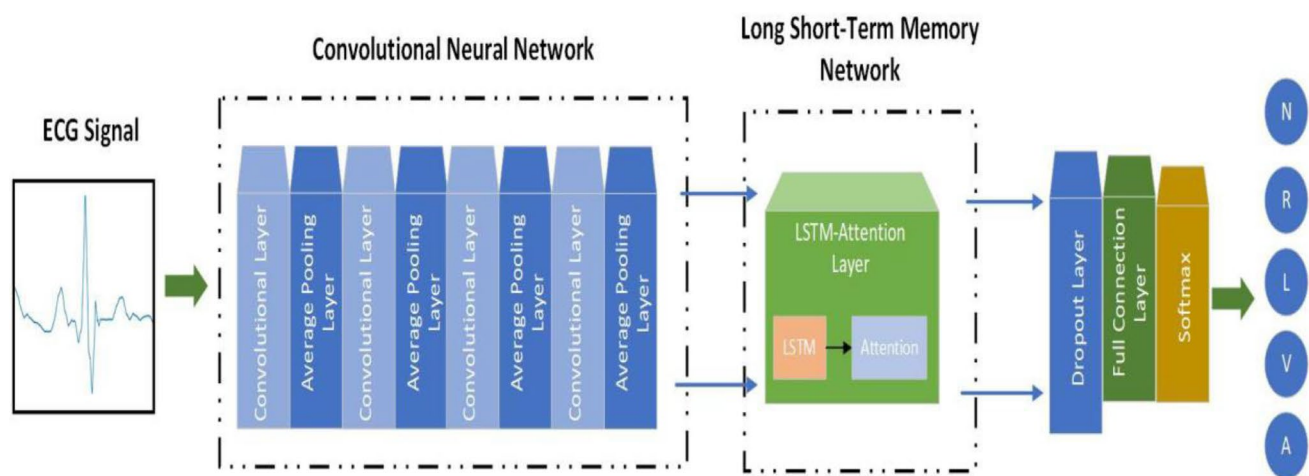


Fig. 2 The comprehensive architecture of the proposed model

alternative strategies such as class weighting and synthetic oversampling (e.g., SMOTE) could also be considered, they pose challenges in maintaining the physiological integrity of ECG signals. SMOTE, for instance, generates synthetic data based on feature interpolation, which may not fully preserve the complex temporal dynamics inherent in real ECG signals. Therefore, the sliding window approach is particularly well-suited for ECG classification, as it expands the dataset while ensuring the biological validity of the generated samples. By employing this targeted data augmentation strategy, the proposed model achieves improved robustness in classifying minority categories, thereby enhancing overall performance in ECG-based arrhythmia detection.

4 Experimental results

4.1 MIT-BIH database

To systematically evaluate and compare the performance of the proposed model, the MIT-BIH Arrhythmia Database (MITDB) was employed as the benchmark dataset. The MITDB consists of 48 half-hour ECG recordings, originally sampled at 360 Hz, with each recording containing two leads [26]. Given that the lead configuration varies across different recordings, 40 records containing lead II and lead V1 were selected to ensure consistency in feature extraction and facilitate direct comparison with prior studies. Following standard preprocessing protocols, ECG signals were segmented based on the position of the R-peak, a fundamental landmark in heartbeat classification. The segmentation strategy involved extracting a 300-sample window, consisting of 100 samples preceding the R-peak and 199 samples following it. This window size was determined to capture the complete morphological structure of each heartbeat while maintaining a fixed-length input for the model. The R-peaks were identified using a robust QRS detection algorithm, ensuring reliable segmentation across different subjects and heartbeat types. In total, 94,131 heartbeats were extracted from the selected 40 recordings. The classification follows the Association for the Advancement of Medical Instrumentation (AAMI) standard, which groups heartbeats into five distinct categories: normal (N), supraventricular (S), ventricular (V), fusion (F), and unknown (Q). The distribution of these categories is detailed in Table 1, where a significant class imbalance is observed, particularly for the S and F categories, necessitating the implementation of data augmentation techniques to mitigate the adverse impact of class imbalance on model performance. By employing a consistent segmentation strategy and adhering to standardized classification protocols, this study ensures a rigorous and reproducible evaluation of the proposed CNN-LSTM model within the MIT-BIH dataset framework.

4.2 Evaluation metrics

In a related study of heartbeat classification. Accuracy, specificity and sensitivity are exploited to evaluate the performance of the model. The confusion matrix represents the detailed information of the classification results of

Table 1 Data distributions of five types of heart beats

Types of heart beats	Quantity
N	7 9607
S	2 702
V	7 135
F	7 93
Q	3 894
In total	9 4131

the model. In multi-class classification problems, the confusion matrix provides a detailed overview of the model's performance by mapping actual versus predicted outcomes. Each row of the matrix corresponds to the true class labels, while the columns represent the predicted class labels. This structure allows for an easy comparison between expected and predicted results, making it a valuable tool for assessing classification accuracy, misclassifications, and the overall reliability of the model. By examining the distribution of errors, it becomes possible to identify specific classes that the model struggles with, which can guide further improvements in model training or feature selection. For K-class multi-classification problems, the confusion matrix is a K-dimensional matrix.

$$\begin{bmatrix} n_{11}^T & \dots & n_{1i}^F & \dots & n_{1K}^F \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ n_{i1}^F & \dots & n_{ii}^T & \dots & n_{iK}^F \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ n_{K1}^F & \dots & n_{Ki}^F & \dots & n_{KK}^T \end{bmatrix}$$

where: n_{ij}^F is the number of samples of class i that are classified as class j , and n_{ii}^T is the number of samples of class i that are correctly classified. P_i is the number of all samples classified as i , and C_i is the number of samples classified as i . For class i , specificity, sensitivity, and accuracy are calculated as follows:

$$Spe = \frac{\sum_{m \neq i} \sum_{l \neq i} n_{ml}}{\sum_{m=1}^K \sum_{l=1}^K n_{ml} - P_i}$$

$$Sen = \frac{n_{ii}^T}{C_i}$$

$$Acc = \frac{n_{ii}^T + \sum_{m \neq i} \sum_{l \neq i} n_{ml}}{\sum_{m=1}^K \sum_{l=1}^K n_{ml}}$$

In the formula, accuracy (Acc) represents the proportion of heartbeats that are correctly classified overall. Specificity (Spe) refers to the proportion of negative examples that are accurately identified, reflecting the model's effectiveness in detecting non-arrhythmic beats. Sensitivity (Sen), on the other hand, denotes the percentage of positive examples that are correctly classified, indicating the model's ability to recognize arrhythmic heartbeats. These metrics together provide a comprehensive assessment of the model's performance in both detecting abnormal conditions and avoiding false positives.

To ensure a rigorous and unbiased evaluation of the proposed CNN-LSTM model, we implemented a systematic data partitioning strategy using the MIT-BIH Arrhythmia Database. From the selected 40 ECG recordings, a total of 94,131 heartbeats were extracted and subsequently divided into two independent sets: approximately 75% (70,598 samples) for model development and 25% (23,533 samples) for final testing. To facilitate effective model training and validation, the development set was further partitioned, with 85% (60,008 samples) allocated for training and 15% (10,590 samples) designated for validation during the training process. This partitioning strategy ensures that test samples remain completely unseen during the model development phase, providing an unbiased assessment of the model's generalization capability.

Fig. 3 The change curves of accuracy in LSTM model

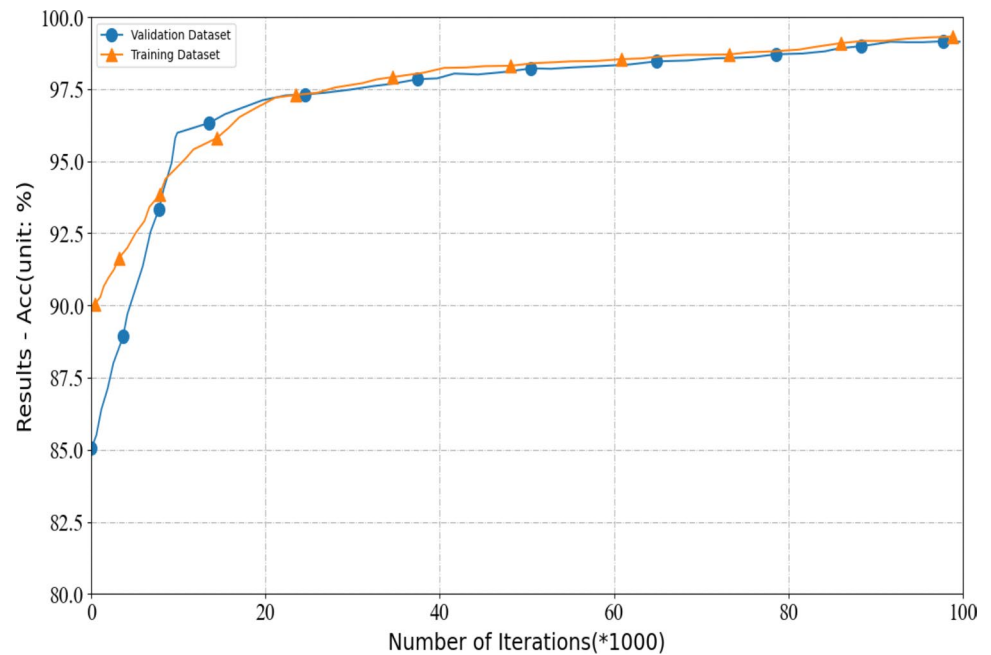
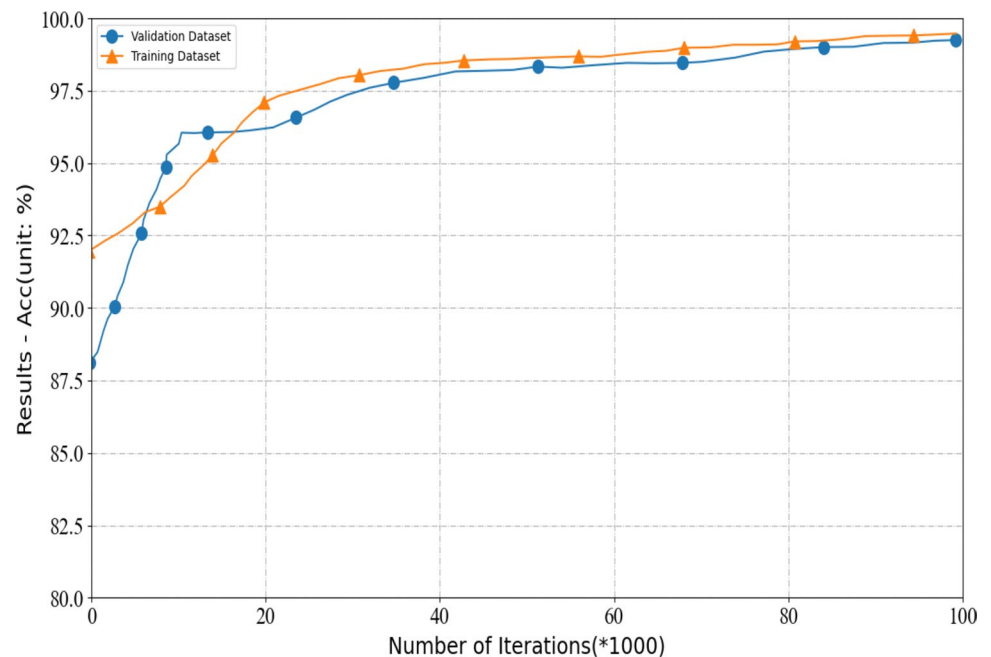


Fig. 4 The change cures of accuracy in CNN-LSTM model



4.3 The results comparison of training and testing with/without CNN

Figures 3 and 4 illustrate the training and validation performance over 10,000 iterations using two different models: one based on the long short-term memory (LSTM) network with raw ECG signals as input, and the other utilizing a convolutional neural network (CNN) for feature extraction. The classification performance of both models is comparable. With raw ECG signals comprising 300 sampling points as input, the LSTM model achieved a peak validation accuracy of 99.16% after 10,000 iterations. In comparison, the CNN-based feature extraction model reached a slightly higher validation accuracy of 99.20% within the same epoch. Neither model exhibited overfitting, demonstrating strong training performance.

However, a significant difference emerged in training time: the LSTM model with raw ECG inputs required approximately 4.5 h, whereas the CNN-LSTM hybrid model reduced the input size to 38 sampling points through feature

extraction, cutting the total training time to just 0.78 h. Both models achieved similar classification accuracy, exceeding 99%, highlighting that the proposed CNN-based feature extraction method effectively reduces the dimensionality of ECG signals. Furthermore, this compression dramatically decreases the training time, thereby lowering the computational cost of the automatic ECG classification process.

4.4 The results comparison of heart beats classification before and after data augmentation

Figure 5 shows the ECG classification results of the parallel combination model of LSTM and CNN. It is worth noting that due to the imbalance in the number of samples of different categories, the model has relatively good recognition ability for N, V, and Q categories, but poor recognition ability for S and F categories. The specificity of class N is low. Representation models tend to classify non-N class beats as N class, i.e., normal beats, which are dangerous in heart disease diagnosis.

As mentioned in data augmentation. In this experiment, the minority class data is sampled by sliding window to suppress the negative impact of sample imbalance on model performance. From Fig. 6, after the parallel combination model of LSTM and CNN is trained on class-balanced data, the sensitivity of class S and class F is increased by 8.57% and 9.38%, respectively. The comparison proves that the sample balancing strategy improves the model's performance.

4.5 Error analysis of misclassified heartbeats

To gain deeper insight into the classification errors made by the proposed CNN-LSTM model, we conducted a detailed error analysis based on the confusion matrix shown in Fig. 7. The confusion matrix, transformed into a heatmap representation with logarithmic scaling to mitigate color disparity due to high numerical differences, provides valuable insights into the model's strengths and weaknesses in ECG heartbeat classification. From the confusion matrix, it is evident that the model demonstrates strong performance in classifying Normal (N) and Ventricular (V) beats, with high true positive counts of 19,720 and 1683, respectively. However, certain misclassifications still occur, particularly in distinguishing supraventricular (S), fusion (F), and unknown (Q) beats, which present lower sample sizes and more complex waveform characteristics. A noticeable number of normal (N) beats are misclassified as S, V, F, and Q, likely due to inherent morphological similarities between normal beats and certain types of arrhythmias, particularly when the signal is slightly distorted due to baseline drift or noise interference during acquisition. The model misclassified 54 instances of N beats as Q beats, suggesting that some normal beats may exhibit abnormal characteristics similar to atrial fibrillation or premature ventricular contractions, causing the classifier to assign them to the Unknown (Q) category. Similarly, Supraventricular (S) beats were misclassified as N or V, indicating a potential overlap in waveform characteristics, particularly when P-wave features are subtle or absent. Ventricular (V) beats were sometimes confused with F and Q beats, as fusion beats inherently exhibit a mixture of normal and ventricular features, contributing to classification ambiguity. Fusion (F) beats,

Fig. 5 Classification results of heart beats types before data augmentation

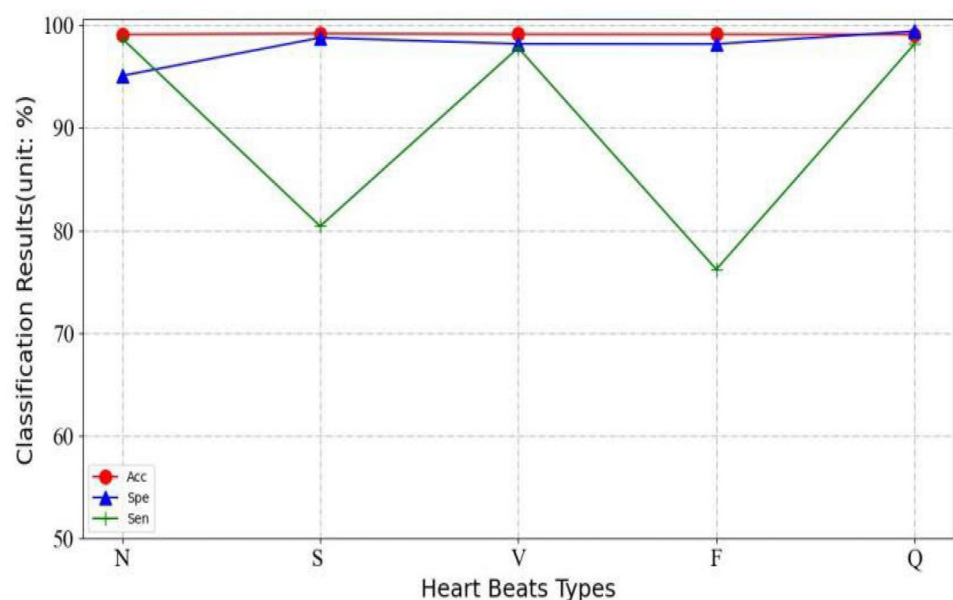


Fig. 6 Classification results of heart beats types after data augmentation

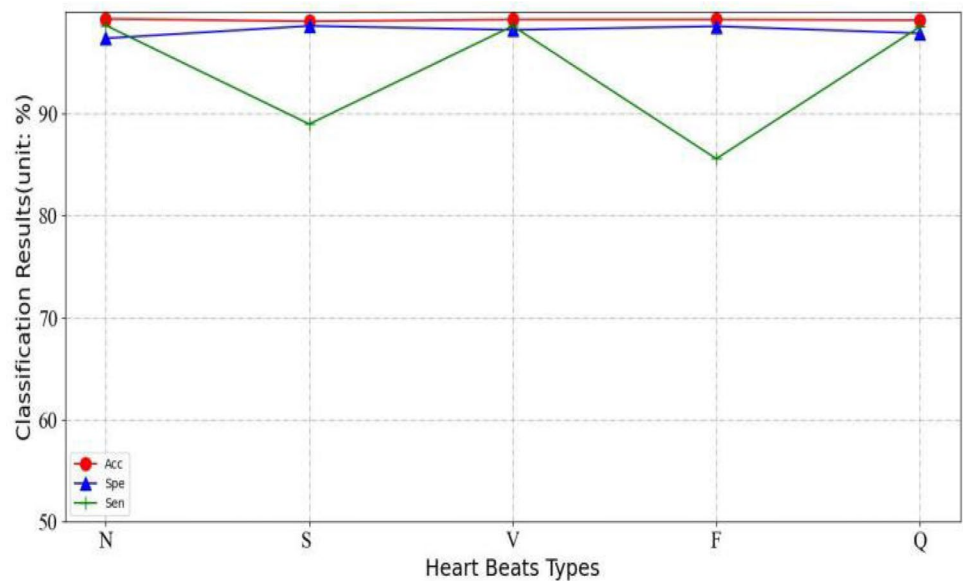
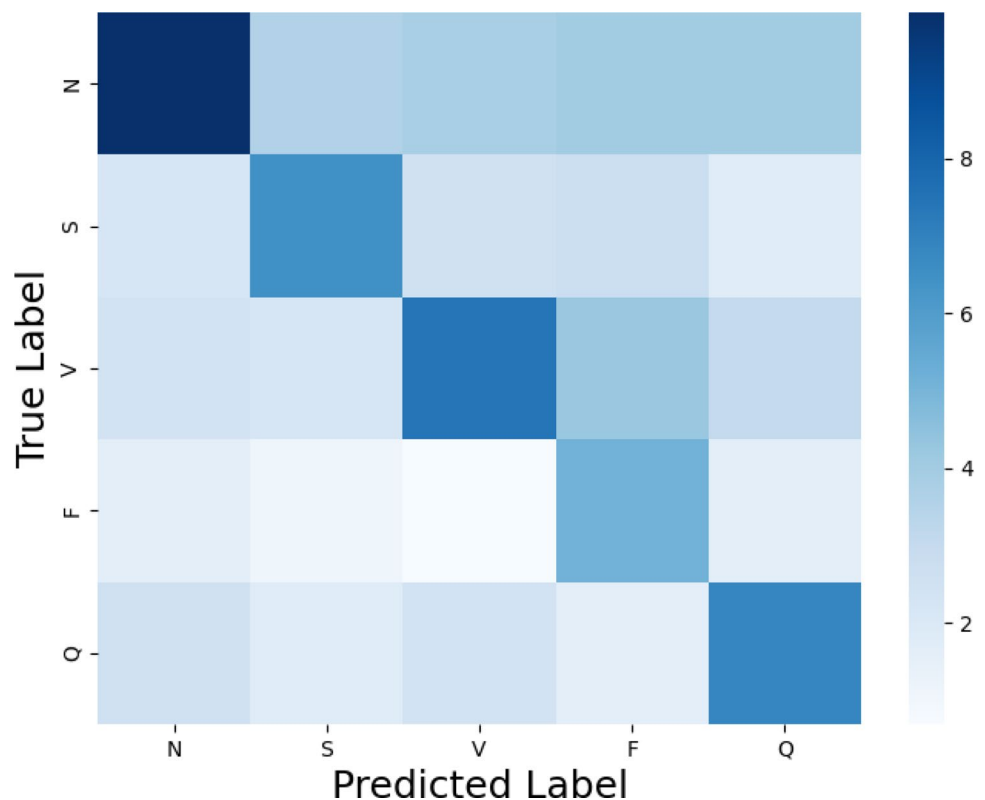


Fig. 7 Confusion matrix heat-map (log-transformed)



which represent hybrid morphologies, were often misclassified as Q beats, reflecting the model's struggle in accurately distinguishing such ambiguous cases. These classification errors highlight challenges related to data imbalance, where minority classes like S and F have fewer training samples, affecting the model's ability to generalize.

Additionally, the morphological similarity among certain arrhythmias complicates classification, and implementing a hybrid approach incorporating lead-wise analysis (e.g., using additional ECG leads like V1 or V5) may improve discrimination. In summary, while the CNN-LSTM model achieves high accuracy in classifying majority classes, further refinements are necessary to improve the recognition of minority classes (S and F) and reduce confusion between morphologically similar arrhythmias. Future work should focus on improving class-balancing strategies, refining feature extraction techniques, and incorporating multi-lead ECG data to enhance overall classification robustness.

4.6 The results comparison of different experimental methods

The comparative analysis in Table 2 highlights the superior performance of our proposed CNN-LSTM model over previous ECG classification methods. While existing models demonstrate competitive accuracy, our approach achieves the highest classification accuracy of 99.20%, outperforming state-of-the-art methods such as Study [33] (1D-RCNN&GWO) and Study [34] (CBAM+AE+MobileNet), which are among the latest advancements in this domain. Notably, our model also maintains a strong balance between sensitivity (94.07%) and specificity (98.14%), ensuring robust detection across different heartbeat categories. The integration of CNN for effective feature extraction and LSTM for capturing long-term dependencies enables our method to surpass traditional machine learning models like SVM and KNN, as well as deep learning architectures that lack sequential modeling. Moreover, by leveraging convolutional operations to reduce ECG signal dimensionality, our model significantly enhances computational efficiency while preserving classification accuracy. These results confirm that our hybrid CNN-LSTM framework provides an optimal balance between accuracy, efficiency, and robustness, making it well-suited for real-time ECG analysis and clinical applications.

4.7 Parameter analysis

Hyperparameter selection is a critical factor influencing the overall performance of deep learning models, particularly in ECG classification tasks where both spatial and temporal dependencies must be effectively captured. In this study, we systematically examine the impact of two key hyperparameters: the number of hidden units in the LSTM layer and the convolutional kernel size in the CNN module. These parameters play a pivotal role in balancing the model's ability to generalize while maintaining computational efficiency.

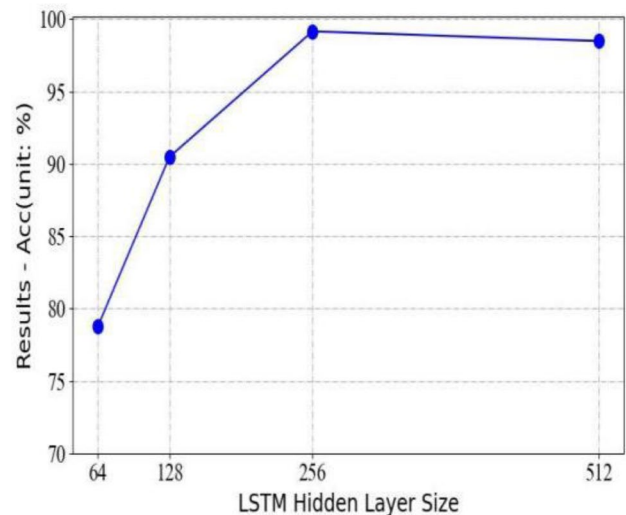
To explore the effect of LSTM hidden layer size, we evaluated four configurations: 64, 128, 256, and 512 units, while keeping other parameters constant. As illustrated in Fig. 7, classification accuracy exhibits a steady increase with a larger hidden layer size, confirming that a greater number of LSTM units enhances the model's capacity to capture long-range temporal dependencies in ECG signals. However, beyond 256 hidden units, the rate of improvement diminishes, and at 512 units, the performance stabilizes without substantial further gains. This suggests that while increasing the hidden layer size allows the model to learn richer temporal features, an excessively large LSTM layer leads to diminishing returns and may introduce unnecessary computational overhead. Consequently, 256 hidden units is identified as the optimal configuration, balancing accuracy and computational efficiency.

Similarly, we investigated the impact of convolution kernel size, testing values ranging from 1 to 10 to determine its influence on feature extraction in the CNN module. As shown in Fig. 8, increasing the kernel size significantly enhances model performance up to a certain threshold. A larger kernel enables the extraction of longer-range spatial features, which is particularly beneficial for ECG signals where waveform morphology carries crucial diagnostic information. However, beyond a kernel size of 7, performance begins to degrade. This decline can be attributed to an overly large receptive field, which may result in the loss of finer local details critical for distinguishing subtle arrhythmia patterns. The observed trend suggests that a kernel size of 7 achieves the best balance, capturing both short-term and long-term morphological patterns in ECG waveforms while avoiding excessive information smoothing.

Table 2 Comparison of arrhythmia classification model performance

Compared methods	Network model	Acc (%)	Sen (%)	Spe (%)
Study [27]	FFNN	96.94	97.78	96.31
Study [28]	KNN	97	97.1	96.9
Study [29]	SVM + RBF	98.91	98.91	97.85
Study [30]	WPE + RF	94.61	–	–
Study [25]	CNN	94.03	96.71	91.54
Study [31]	MWT + SVM	98.39	96.86	98.92
Study [32]	CNN	98.42	72.06	97.83
Study [33]	1D-RCNN&GWO	98.20	–	–
Study [34]	CBAM+AE+MobileNet	98.21	95.03	–
Our method	CNN+LSTM	99.20	94.07	98.14

Fig. 8 Influence of LSTM hidden layer size



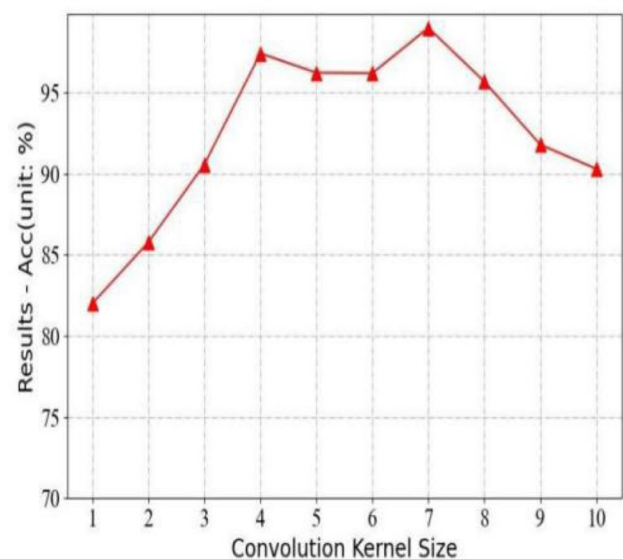
These findings underscore the importance of careful hyperparameter tuning to optimize ECG classification performance. The results suggest that while increasing model complexity can enhance performance, excessive parameter growth does not necessarily yield proportional improvements and may introduce computational inefficiencies. By setting 256 LSTM hidden units and a convolutional kernel size of 7, we achieve an optimal trade-off between classification accuracy and model efficiency, making our approach well-suited for real-time IoMT applications (see Fig. 9).

4.8 Hardware complexity analysis

The computational efficiency of our proposed model stems from two key architectural designs. The CNN module effectively compresses the 300-point ECG signal into a 38-dimensional feature vector, reducing the input dimension by 87.3%. This dimensional reduction significantly decreases the computational complexity for the subsequent LSTM processing. Furthermore, our hybrid architecture separates feature extraction and temporal analysis, allowing for efficient parallel processing and memory utilization.

From a theoretical perspective, the computational complexity of our model can be analyzed in two stages. For the CNN feature extraction stage, the complexity is $O(k \cdot n \cdot c)$, where k is the kernel size (7 in our implementation), n is the input length (300), and c is the number of channels. The LSTM processing stage has a complexity of $O(h \cdot m)$, where h is the hidden size (256) and m is the compressed feature length (38). Compared to processing raw ECG signals directly with

Fig. 9 Influence of convolution kernel layer size



LSTM which has a complexity of $O(h \cdot n)$, our approach reduces the temporal sequence length from 300 to 38, leading to approximately $7.9 \times$ reduction in LSTM computational overhead while maintaining classification accuracy.

To validate the practical benefits of our design, we implemented the model on NVIDIA Jetson Nano platform equipped with 4 GB RAM and quad-core ARM Cortex-A57 processor. The experimental results show that our model achieves an inference time of 85 ms per heartbeat segment. The results demonstrate that our model achieves real-time performance with significant margin. The CNN-based feature compression not only accelerates computation but also enables efficient hardware utilization while maintaining high classification accuracy.

5 Conclusions

In term of IoMT, the integration of sports and medical care is a hot research topic today. The use of wearable devices to monitor physiological indicators is of great benefit to both ordinary people and professional athletes participating in sports. ECG signals can be used to reflect the state of the heart. This study presents a CNN-LSTM hybrid model for ECG classification, achieving 99.20% accuracy, 94.07% sensitivity, and 98.14% specificity on the MIT-BIH Arrhythmia Database. The model effectively captures morphological and temporal dependencies in ECG signals while reducing computational overhead by $7.9 \times$ compared to LSTM-only approaches. Real-time feasibility is confirmed with 85 ms inference time on an NVIDIA Jetson Nano, making it suitable for IoMT applications. Future work will focus on integrating multi-lead ECG signals and improving classification of minority heartbeat classes to enhance clinical applicability.

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Data availability The data will be made available upon reasonable request.

Declarations

Ethics approval and consent to participate Not applicable.

Consent for publication Not applicable.

Competing interests The authors declare no competing interests.

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